

Multi-VPC Connectivity and Transit Gateway

A wire, a router, or a doorway. Three answers, and they are not interchangeable.

Peering is a **wire**. Transit Gateway is a **router**. PrivateLink is a **doorway** to one service.

Choosing between the three

APPROACH	REACH FOR IT WHEN	DEAL BREAKER
VPC peering	Two VPCs, high volume, no transit needed	Not transitive; CIDRs must not overlap
Transit Gateway	Many VPCs, hybrid, segmentation needed	Per-GB processing charge on everything
PrivateLink	Expose one service, keep networks separate	One service only; needs an NLB or GWLB

ON THE EXAM

Overlapping CIDRs eliminate peering and Transit Gateway together. Only PrivateLink is indifferent to the address space.

VPC peering: four limits that decide questions

- ◆ **Not transitive.** If A peers with B and B with C, A still cannot reach C.
- ◆ **No edge-to-edge routing.** You cannot use a peer's internet gateway, NAT device, VPN, Direct Connect or endpoints.
- ◆ **CIDRs must not overlap,** and only one peering connection can exist between the same two VPCs.
- ◆ **50 active connections per VPC** by default, raisable to 125. Cross-region and cross-account both work.
- ◆ **No per-GB processing charge.** This is peering's real advantage and the exam does test it.

ON THE EXAM

Edge-to-edge is the forgotten one. Routing your traffic out through a peer's NAT gateway simply does not work.

Ten VPCs fully meshed is 45 connections.
Fifty VPCs is **1,225**, each with route table
entries on both sides.

Transit Gateway fundamentals

- ◆ **Regional.** Attach VPCs, VPNs, Direct Connect gateways, and other Transit Gateways.
- ◆ **Route tables do the work.** One TGW can hold many, and that is how segmentation happens.
- ◆ **5,000 attachments** per gateway. One VPC attachment per VPC per gateway; up to 5 gateways per VPC.
- ◆ **50 Gbps burst** per VPC attachment. MTU 8500 inside, but only 1500 over VPN.
- ◆ Share it across accounts with **RAM**, exactly as in module five. Do not build one per account.

ON THE EXAM

The bandwidth ceiling is per attachment, not per gateway. A single very hungry VPC is the case where it bites.

Follow the packet through a Transit Gateway

01

Subnet route

The subnet route table sends the packet to the TGW attachment.

02

Association

The attachment is associated with exactly one TGW route table. That one is consulted.

03

Lookup

That table decides which attachment the packet leaves by.

04

Delivery

The destination VPC's own subnet route table takes over.

ON THE EXAM

Association picks which table an attachment reads. Propagation decides which tables learn about it. They are different questions.

Association or propagation?

ASSOCIATION

Which table I read

- Exactly one table per attachment
- Decides where my traffic may go
- Controls what this VPC can reach
- Change it to move a VPC's segment

PROPAGATION

Which tables learn about me

- Can propagate into many tables
- Decides who can reach me
- Controls this VPC's visibility
- Or add static routes by hand

THE TIEBREAKER

Segmentation is built by controlling propagation. Spokes associate with a table holding only shared services routes.

Appliance mode, and why flows break without it

- ◆ By default a TGW keeps traffic in the Availability Zone it arrived in, choosing per direction.
- ◆ With a firewall in the path, the return packet can land on a different appliance, which has no record of the flow, and drops it.
- ◆ **Appliance mode** pins both directions of a flow to the same AZ, so the same appliance sees both.
- ◆ Enable it on the **inspection VPC attachment**. Symptom: intermittent failures that look like packet loss.

ON THE EXAM

Stateful appliance plus multiple AZs plus mysterious asymmetric drops equals appliance mode, every time.

Newer capabilities worth knowing

- ◆ **Security group referencing across a TGW**, since late 2024. Reference a group in another attached VPC.
- ◆ It does **not** work across TGW peering connections. Both VPCs must attach to the same gateway.
- ◆ **Network function attachments** connect a TGW straight to AWS Network Firewall, with no inspection VPC.
- ◆ **Encryption support** enforces encryption in transit for all TGW traffic.

ON THE EXAM

Security group referencing removed the main reason teams stayed on peering. Older study material still says it is impossible.

The cost model, because it decides answers

- ◆ **TGW charges twice:** an hourly fee per attachment, and a charge per GB processed.
- ◆ **Peering charges no processing fee at all.** You pay only normal data transfer.
- ◆ So for two VPCs moving very large volumes, peering is meaningfully cheaper.
- ◆ For many VPCs, the TGW fee buys away the mesh, and usually wins on both cost and operations.

ON THE EXAM

Peering is cheaper per byte. Transit Gateway is cheaper per relationship. The VPC count tells you which one the question is about.

Reading the scenario

THE SCENARIO SAYS

Dev must not reach Prod, both reach shared services

Intermittent drops through a firewall across AZs

Overlapping CIDRs, one service to expose

Two VPCs, enormous traffic, nothing else needed

VPC count keeps growing, plus on-premises

Route out through the other VPC's NAT gateway

IT IS POINTING AT

TGW route table segmentation

Appliance mode

PrivateLink

VPC peering

Transit Gateway

Impossible over peering

ON THE EXAM

Count the VPCs first, then check the CIDRs. Those two facts decide most of these before you read the options.

Meridian Freight has 14 VPCs connected by a partial mesh of peering connections. Two new VPCs are planned each quarter. Every VPC also needs to reach the on-premises data centre, currently through a VPN terminating in one VPC.

Teams report that some VPC pairs cannot reach each other, and that on-premises traffic works from only one VPC.

What should the solutions architect recommend?

- ☐ A Complete the mesh so every VPC peers with every other VPC, and add a VPN in each VPC
- ☐ B Deploy a Transit Gateway, attach every VPC, and move the VPN onto it
- ☐ C Configure transitive routing on the existing peering connections
- ☐ D Route all inter-VPC traffic through the VPC that terminates the VPN

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Complete the mesh
Ninety-one connections today, growing every quarter, plus sixteen VPN connections to pay for and maintain. It scales in the wrong direction.
- B

 Transit Gateway
One attachment per VPC, one VPN, and a new VPC is one attachment rather than fourteen peerings.
- C

 Enable transitive routing
There is no such setting. Peering is non-transitive by design, and no configuration changes that.
- D

 Hub through the VPN VPC
This is exactly what non-transitive means. Traffic cannot pass through a peered VPC to reach a third.

THE SIGNAL

Both reported symptoms are the same limitation. Non-transitivity explains the unreachable pairs and the single-VPC on-premises access.

A company runs Production, Development and Shared Services VPCs attached to one Transit Gateway. Security requires that Development can never reach Production, while both must reach Shared Services, and Shared Services must reach both.

The current design uses one Transit Gateway route table with all attachments propagating into it.

How should the architect meet the requirement?

- ☐ A Add network ACLs in the Production VPC denying the Development CIDR
- ☐ B Use separate TGW route tables: spokes associate with a table holding only Shared Services routes
- ☐ C Detach Development and connect it by VPC peering instead
- ☐ D Apply an SCP denying network access between the accounts

◆ PAUSE AND DECIDE

How should the architect meet the requirement?

- A** Network ACLs
Works, and depends on a deny list staying correct as CIDRs change. The route still exists, so a mistake restores reachability silently.
- B** Separate TGW route tables
The spoke table contains no route to Production, so the path does not exist. Nothing to misconfigure later.
- C** Peering for Development
Solves isolation by leaving the design. Development would then lose Shared Services and on-premises reachability.
- D** An SCP
SCPs govern API calls, not packets. No policy in Organizations can stop one instance reaching another.

THE SIGNAL

Removing the route beats denying the traffic. If the path does not exist, there is no rule for anyone to get wrong.

An inspection VPC runs a fleet of stateful third-party firewalls behind a Gateway Load Balancer, spread across three Availability Zones, with all inter-VPC traffic routed through it by the Transit Gateway.

Long-lived connections fail intermittently. Short requests usually succeed. Firewall logs show packets arriving for sessions the appliance has no record of.

What is the cause?

- ☐ A The Gateway Load Balancer is unhealthy and needs deregistration delay tuning
- ☐ B Appliance mode is not enabled on the inspection VPC attachment
- ☐ C The firewall instances need larger instance types
- ☐ D MTU mismatch is fragmenting packets between VPCs

◆ PAUSE AND DECIDE

What is the cause?

A Load balancer health

An unhealthy appliance fails every flow sent to it, not a subset. It also would not produce sessions with no record.

C Bigger instances

A capacity answer to a symptom that is not about capacity. Load is never mentioned.

B Appliance mode not enabled

Without it the TGW picks an AZ per direction, so the return packet reaches a different appliance that never saw the session.

D MTU mismatch

Real and worth knowing, since TGW is 8500 and VPN is 1500. It would affect large packets, not whole sessions the appliance does not recognise.

THE SIGNAL

Packets arriving for unknown sessions is the giveaway. That is asymmetry, and asymmetry across AZs is what appliance mode fixes.

Following an acquisition, the parent company must let the acquired company's applications call an internal payments API. Both estates use 10.0.0.0/16 with heavily overlapping subnets, and renumbering either is out of the question for at least two years.

Only that one API needs to be reachable.

What should the solutions architect recommend?

- A VPC peering between the two environments
- B Attach both estates to a Transit Gateway
- C Expose the API through PrivateLink, with an endpoint service behind a Network Load Balancer
- D A Site-to-Site VPN between the two VPCs

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 VPC peering
Refused outright. Peering requires non-overlapping CIDRs, so the connection cannot even be created.
- B

 Transit Gateway
Also routing, so it needs unambiguous addresses. Overlapping ranges cannot be routed between.
- C

 PrivateLink endpoint service
Not routing at all. The consumer reaches a local endpoint IP, so the two address spaces never meet.
- D

 Site-to-Site VPN
Routing again, with the same overlap problem, plus tunnel cost and complexity for a single API.

THE SIGNAL

Overlapping CIDRs eliminate every routing answer at once. PrivateLink is the only option that never routes between the two networks.

A team is migrating from a peering mesh to a Transit Gateway. Their security groups rely heavily on referencing security groups in peered VPCs rather than CIDR blocks.

After cutting the first two VPCs over, connectivity fails. An engineer proposes rewriting every rule to use CIDR blocks.

What should the solutions architect advise?

- ☐ A Enable security group referencing on the Transit Gateway and the VPC attachments
- ☐ B Rewrite the rules to CIDR blocks, as referencing is unsupported over a Transit Gateway
- ☐ C Keep peering alongside the Transit Gateway purely for the security group references
- ☐ D Replace security groups with network ACLs referencing the peer VPC

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

Enable SG referencing on the TGW
Supported since late 2024 for VPCs attached to the same gateway. It is a setting, not a rewrite.
- B

Rewrite to CIDRs
Was correct before late 2024 and is now outdated. It also discards the maintainability the references were giving them.
- C

Keep peering too
Retains the mesh they are migrating away from, and doubles the paths to reason about during troubleshooting.
- D

Network ACLs
NACLs are stateless and only understand CIDR ranges. They cannot reference a security group at all.

THE SIGNAL

This answer changed. Material written before late 2024 says option B, and it was right at the time.

A media company runs a rendering VPC and a storage VPC in the same Region. They exchange roughly 900 TB per month and need nothing else on the network. The team has attached both to the company Transit Gateway.

Finance has queried a large and growing Transit Gateway line on the bill.

What should the solutions architect recommend?

- ☐ A Leave it; Transit Gateway is the AWS-recommended pattern for VPC connectivity
- ☐ B Add a VPC peering connection between the two VPCs and route that traffic over it
- ☐ C Move both workloads into a single VPC
- ☐ D Enable a Transit Gateway multicast domain to reduce data processing

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Leave it as is
Pays a per-GB processing charge on 900 TB every month to solve a problem that has only two endpoints.
- B

 Peer the two VPCs
Peering carries no processing charge. Two VPCs needing nothing else is exactly the case where the mesh objection does not apply.
- C

 Merge into one VPC
Would remove the charge and also the isolation, quotas and blast-radius separation the two VPCs provide.
- D

 Multicast domain
Multicast is for one-to-many distribution. It has no bearing on point-to-point volume or on processing charges.

THE SIGNAL

Pair this with scenario one. Fourteen growing VPCs made Transit Gateway right. Two static VPCs and huge volume reverses it.

What to carry out of this module

- ◆ **Wire, router, doorway.** Peering, Transit Gateway, PrivateLink. Name the shape the scenario needs.
- ◆ **Overlapping CIDRs eliminate every routing answer.** Only PrivateLink is indifferent to addresses.
- ◆ **Association is which table I read. Propagation is which tables learn about me.** Segmentation is built from propagation.
- ◆ Stateful appliances across AZs need **appliance mode**. Unknown-session drops are the signature.
- ◆ Peering is cheaper per byte. Transit Gateway is cheaper per relationship.

ON THE EXAM

Count the VPCs, then check the CIDRs. Those two facts decide the majority of questions in this area before you read the options.